

Experiment 4

E-Commerce App using React Router

App.js

```
// Program 4: Simplified e-commerce app with React Router (works fully offline)
// Needs: npm install react-router-dom (one-time; afterwards no internet needed)
// Usage: save as src/App.jsx in a Vite/CRA project.
import React, { createContext, useContext, useState } from "react";
import {
  BrowserRouter, Routes, Route, Link, NavLink,
  Navigate, useParams, useNavigate, useLocation,
} from "react-router-dom";

// ----- Local data (no API) -----
const PRODUCTS = [
  { id: 1, name: "Laptop", price: 55000, desc: "14-inch, 16GB RAM, 512GB SSD" },
  { id: 2, name: "Headphones", price: 2500, desc: "Wireless, noise cancelling" },
  { id: 3, name: "Smart Watch", price: 4000, desc: "Heart-rate and sleep tracking" },
  { id: 4, name: "Backpack", price: 1200, desc: "Water-resistant, 25L" },
];

// ----- Global state (auth + cart) -----
const AppContext = createContext();
const useApp = () => useContext(AppContext);

function AppProvider({ children }) {
  const [user, setUser] = useState(null);
  const [cart, setCart] = useState([]);

  const addToCart = (product) => setCart((c) => [...c, product]);
  const removeFromCart = (index) => setCart((c) => c.filter((_, i) => i !== index));

  return (
    <AppContext.Provider value={{ user, setUser, cart, addToCart, removeFromCart }}>
      {children}
    </AppContext.Provider>
  );
}

// ----- Protected route -----
function ProtectedRoute({ children }) {
  const { user } = useApp();
  const location = useLocation();
  if (!user) return <Navigate to="/login" state={{ from: location.pathname }} replace />;
  return children;
}

// ----- Navigation -----
function Navbar() {
  const { user, setUser, cart } = useApp();
  const navigate = useNavigate();
  const style = ({ isActive }) => ({
    marginRight: 15, textDecoration: "none",
    fontWeight: isActive ? "bold" : "normal", color: isActive ? "#c0392b" : "#333",
  });

  return (
    <nav style={{ padding: 12, background: "#eee", display: "flex", alignItems: "center" }}>
      <NavLink to="/" style={style}>Home</NavLink>
      <NavLink to="/products" style={style}>Products</NavLink>
      <NavLink to="/contact" style={style}>Contact</NavLink>
      <NavLink to="/cart" style={style}>Cart ({cart.length})</NavLink>
      <span style={{ marginLeft: "auto" }}>
        {user ? (
          <>
            Hi, {user}{" "}
            <button onClick={() => { setUser(null); navigate("/"); }}>Logout</button>
          </>
        ) : (
          <NavLink to="/login" style={style}>Login</NavLink>
        )}
      </span>
    </nav>
  );
}

// ----- Pages -----
const Home = () => (
```

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App.js (Contd.)

```
<div>
  <h2>Welcome to MyShop</h2>
  <p>Browse our products and add your favourites to the cart.</p>
  <Link to="/products">Shop now</Link>
</div>
);

const Products = () => (
  <div>
    <h2>Products</h2>
    {PRODUCTS.map((p) => (
      <div key={p.id} style={{ border: "1px solid #ccc", padding: 10, margin: "10px 0" }}>
        <h3>{p.name}</h3>
        <p>■{p.price}</p>
        <Link to={`/${products}/${p.id}`}>View details</Link>
      </div>
    ))}
  </div>
);

function ProductDetails() {
  const { id } = useParams(); // dynamic routing
  const { addToCart } = useApp();
  const product = PRODUCTS.find((p) => p.id === Number(id));

  if (!product) return <h2>Product not found</h2>;
  return (
    <div>
      <h2>{product.name}</h2>
      <p>{product.desc}</p>
      <p><b>■{product.price}</b></p>
      <button onClick={() => addToCart(product)}>Add to cart</button>{" "}
      <Link to="/products">Back</Link>
    </div>
  );
}

function Contact() {
  const [sent, setSent] = useState(false);
  return (
    <div>
      <h2>Contact Us</h2>
      {sent ? (
        <p>Thanks! We will get back to you.</p>
      ) : (
        <input placeholder="Your email" /><br /><br />
        <textarea placeholder="Message" /><br /><br />
        <button onClick={() => setSent(true)}>Send</button>
      )}
    </div>
  );
}

function Login() {
  const { setUser } = useApp();
  const navigate = useNavigate();
  const location = useLocation();
  const [name, setName] = useState("");
  const [pass, setPass] = useState("");
  const [error, setError] = useState("");

  const handleLogin = () => {
    // Demo credentials: any name with password "1234"
    if (name && pass === "1234") {
      setUser(name);
      navigate(location.state?.from || "/", { replace: true });
    } else {
      setError("Enter a username and the password 1234");
    }
  };

  return (
```

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App.js (Contd.)

```
<div>
  <h2>Login</h2>
  <input placeholder="Username" value={name} onChange={(e) => setName(e.target.value)} /><br /><br />
  <input type="password" placeholder="Password (1234)" value={pass}
    onChange={(e) => setPass(e.target.value)} /><br /><br />
  <button onClick={handleLogin}>Login</button>
  {error && <p style={{ color: "red" }}>{error}</p>}
</div>
);
}

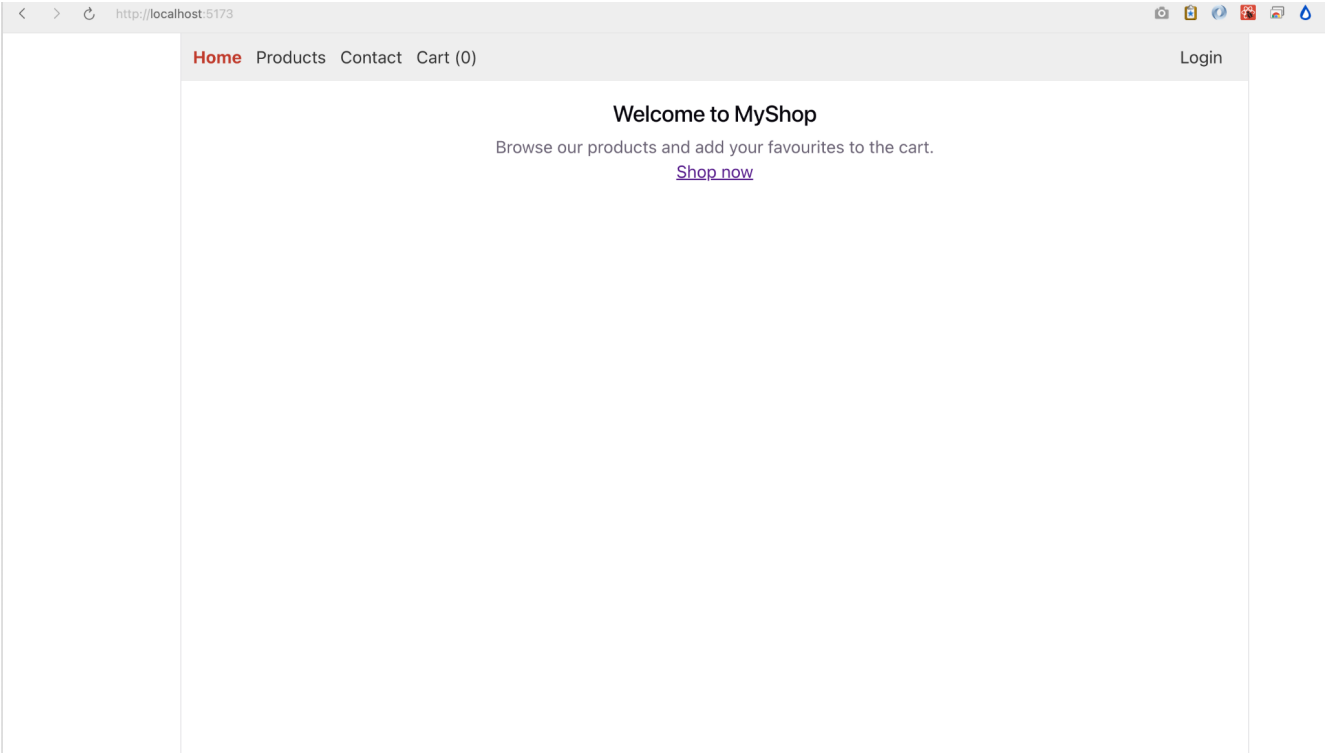
function Cart() {
  const { cart, removeFromCart } = useApp();
  const total = cart.reduce((sum, p) => sum + p.price, 0);
  return (
    <div>
      <h2>Your Cart</h2>
      {cart.length === 0 ? <p>Cart is empty.</p> : (
        <div>
          {cart.map((p, i) => (
            <p key={i}>
              {p.name} - ■{p.price}{ " "}
              <button onClick={() => removeFromCart(i)}>Remove</button>
            </p>
          ))}
          <h3>Total: ■{total}</h3>
        </div>
      )}
    </div>
  );
}

const NotFound = () => (
  <div>
    <h2>404 - Page Not Found</h2>
    <Link to="/">Go back home</Link>
  </div>
);

// ----- App -----
export default function App() {
  return (
    <AppProvider>
      <BrowserRouter>
        <Navbar />
        <div style={{ padding: 20 }}>
          <Routes>
            <Route path="/" element={<Home />} />
            <Route path="/products" element={<Products />} />
            <Route path="/products/:id" element={<ProductDetails />} />
            <Route path="/contact" element={<Contact />} />
            <Route path="/login" element={<Login />} />
            <Route path="/cart" element={<ProtectedRoute><Cart /></ProtectedRoute>} />
            <Route path="*" element={<NotFound />} />
          </Routes>
        </div>
      </BrowserRouter>
    </AppProvider>
  );
}
```

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Output



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E-Commerce App using React Router

Aim

1. To build a simplified e-commerce application using React Router.
2. To implement dynamic routing, protected routes and navigation links.

Procedure

1. Create the React application and install react-router-dom.
2. Create Home, Products, Product Details, Contact, Login, Cart and 404 pages.
3. Configure navigation, dynamic product routes and protected Cart access.
4. Run the application and verify all routes.

Sample Input

- Navigate through Home, Products, Contact, Login and Cart; open a product details page.

Sample Output

- Home, Products, Product Details, Contact, Login, Cart and 404 pages work through React Router.

Result

Thus, the simplified e-commerce application was successfully implemented using React Router.